

ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI - NMTC - 2023 Jan. - SUB-JUNIOR - FINAL

Instructions:

1. Answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your 'rough work' should be in the answer sheet itself.
6. Maximum time allowed is THREE hours

1. A tray contains of 40 toffees. Jaya and Uma take in turn some toffees from the tray. Each time they are allowed to take 1, 2 or 3 toffees only. The person who gets the last toffee, wins. Jaya starts the game. Will she win? If so, how? If not, why?

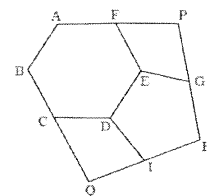
2. Given $A = \left\{ \frac{\frac{(a+b)^2 + (a-b)^2}{b-a} - (a+b)}{\frac{1}{b-a} - \frac{1}{a+b}} \right\} \div \left\{ \frac{(a+b)^3 + (b-a)^3}{(a+b)^2 + (a-b)^2} \right\}$

$B = \left\{ \frac{c-b}{(a-b)(a-c)} - \frac{c-a}{(b-c)(b-a)} + \frac{b-a}{(c-a)(a-b)} \right\} \div \frac{2(a^2 + b^2 + c^2 - ab - bc - ca)}{(a-b)(b-c)(c-a)}$

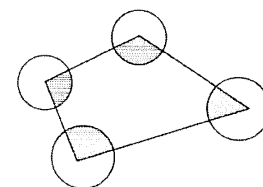
If $a = 2022$, $b = 2023$, find $(A+B)$.

3. There are square papers of areas 1, 2, 3, . . . square millimetres. Asif started colouring them with red and green paint. He painted in red the papers whose areas can be written as the sum of two composite numbers and the rest in green. How many papers are painted green and what are their areas?
4. Find natural numbers a , b , c such that their sum is 6, sum of their squares is 14 and the sum of the products of a and c , and b and c is equal to the square of one more than the product of a and b .
5. The volumes of three cubic containers C_1 , C_2 , and C_3 are in the ration 1: 8: 27. The amount of water in them are in the ratio 1:2:3. Water is poured from C_1 to C_2 and from C_2 to C_3 then the water level in all the containers is the same. Now $128\frac{4}{7}$ litres of water is poured out from C_3 to C_2 after which a certain amount is poured from C_2 to C_1 so that the depth of water in C_1 becomes twice that in C_2 . This results in the amount of water in C_1 100 litres less than the original amount. How much water did each container contain originally?

6. In the adjoining figure, $ABCDEF$ and $DEGHI$ are respectively a regular hexagon and a regular pentagon. P is the intersection of AF and HG and similarly Q . The bisector of $\angle FPG$ cuts AB produced at R . Prove that $\angle Q = 8\angle R$.



7. PQR is an equilateral triangle. S is any point inside the triangle. SA , SB , SC are respectively drawn perpendiculars to PR , RQ and PQ . Find the ratio of $\frac{SA+SB+SC}{QA+RB+PC}$.



8. In the adjoining figure, four equal circles of radius 7 cm each are drawn with centres at the four vertices of a quadrilateral. Find the total area of the shaded sectors.

End of Question Paper